

# The Geometry of Computation

## Topological Gaps, Coordinate Inversions, and the Resolution of Complexity Paradigms

Driven by Dean Kulik

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### Introduction: The Dielectric Geometry of Information

Classical computational theory has long been predicated on the temporal execution of sequential logic. From the earliest formalized Turing machines to modern von Neumann architectures, computation is universally modeled as an iterative, chronological evolution of states:  $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \dots \rightarrow S_k$ . However, a rigorous mathematical and physical analysis reveals that this sequential "forward stream" is merely an artifact of the coordinate system chosen by an external observer. When computation is reframed as a geometric and topological structure, the concepts of time, algorithmic delay, and computational complexity cease to be fundamental physical laws. Instead, they emerge as projections of spatial coupling.

The foundational axiom of this framework requires examining the boundary conditions between two active processes—specifically, a reading process and a writing process. In any computational or physical system, a dielectric gap—a structural delay or separation—is geometrically required only when a reader and a writer are topologically "facing" one another, or when one is nested within the other. Two processes running in parallel without information exchange require no physical or temporal separation; they merely coexist without interaction. But the moment these processes couple—when the write mechanism directly feeds the read mechanism—a dielectric gap becomes structurally mandatory to prevent a degenerate short circuit.

This short circuit is the mathematical equivalent of evaluating  $x = f(x)$  in zero time, yielding an infinite, unresolvable loop that computes nothing. In nested systems, where an internal computational model operates inside the external environment it seeks to represent, the dielectric gap is the sole boundary maintaining the distinction between the model and the modeled reality. Without this gap, the internal representation collapses into the external world, losing its capacity to compute, abstract, or predict. Therefore, the "lag" or "latency" observed in classical computation is not fundamentally a temporal constraint. It is a relational constraint. The gap is a geometric necessity of the coupling. If the topological arrangement of the system is altered so that the components no longer face each other, the necessity for the dielectric vanishes, and the computational distance collapses.

This report presents an exhaustive theoretical synthesis that maps the classical "forward stream" of sequential execution against the "side stream" of direct-address computation. By anchoring this analysis in the Bailey-Borwein-Plouffe (BBP) formula for  $\pi$ , tridiagonal Jacobi matrices, orthogonal similarity transforms, Geometric Complexity Theory, and the physics of delay-coupled oscillators, this treatise demonstrates that algorithmic complexity—and specifically the  $P$  versus  $NP$  problem—is an expression of coordinate topology. Hardness in computation does not arise from the sheer volume of a search space, but rather from the topological impossibility of incrementally navigating from a coupled basis to a diagonalized one.

## The Physical Topology of the Gap: Dielectrics and Delay Systems

To understand the geometric necessity of the gap in abstract computation, it is necessary to first examine how this gap physically manifests in electromagnetic and dynamical systems. The physical world rigorously enforces the dielectric gap to preserve the integrity of interacting states.

### Transmission Lines and Dielectric Propagation

In high-speed printed circuit boards (PCBs) and digital data transmission lines, signals do not propagate instantaneously. The transmission line propagation delay ( $t_{PD}$ ) is the finite duration of time a data signal takes to travel from the source (the writer) to the load (the reader).

This delay is intrinsically tied to the characteristic impedance ( $Z_0$ ) of the line, which is defined by the distributed inductance ( $L_0$ ) and capacitance ( $C_0$ ). The propagation delay per unit length is defined precisely as the square root of the inductance-capacitance product:

$$t_{PD} = \sqrt{L_0 C_0}$$

The physical separation between the conductive traces is filled with a dielectric material. The dielectric constant ( $E_r$ , or relative permittivity) governs the speed at which electromagnetic signals propagate, acting as a frictional medium that slows the signal relative to the speed of light in a vacuum ( $c = 3 \times 10^8$  m/s).

In this physical architecture, the driver and the receiver face each other across the conductive trace. The dielectric material is the physical instantiation of the geometric gap. Without this gap—if the impedance is severely mismatched, or if the dielectric material is fundamentally inconsistent—the signal suffers massive reflections, parasitic stray capacitance, and data skew.

The mechanical equivalent of this phenomenon is wave propagation along a string with varying mass densities. If an impulse travels along a string with density  $\rho$  and hits a boundary connected to a string of density  $\alpha$ , the behavior is entirely dictated by the impedance matching at the boundary. If  $\alpha = \infty$  (representing a short circuit or an immovable boundary), there is full reflection with the signal inverted. If the impedance is perfectly matched, the wave passes through with no reflection. The dielectric gap is precisely what prevents the read/write pair from degenerating into a chaotic short circuit.

### Delay-Coupled Oscillators and Neural Feedback

The geometric requirement for a gap is even more pronounced in dynamical systems with feedback, such as biological neural networks, genetic regulatory circuits, and physical oscillatory rings. A single oscillator with delayed self-feedback or a ring of feed-forward oscillators must manage the finite velocity of signal propagation to maintain coherent states.

In the case of delayed nonlinear feedback, the system cannot be modeled with standard ordinary differential equations. It is governed by a delayed differential equation (DDE), such as:

$$\frac{dx_i}{dt} = g(x_i) + \sum_{j \neq i} A_{ij} f(x_j(t - \tau_j))$$

where  $\tau_j > 0$  represents the absolute time delay for the signal propagating from node  $j$  to node  $i$ .

This  $\tau$  parameter serves as the computational dielectric. In complex systems like the Hodgkin-Huxley neuronal model or networks of delayed coupled Kuramoto oscillators, the delay  $\tau$  is strictly necessary to prevent the system from losing its temporal landmarks. If  $\tau = 0$  in a strong negative feedback loop, the system attempts to evaluate its future state simultaneously with its current state—resulting in catastrophic instability.

When the dielectric gap  $\tau$  is properly tuned, these systems exhibit remarkable emergent complexity, generating families of rotating waves, multi-jittering solutions, and stable periodic orbits. The number of stable periodic orbits often scales linearly with the roundtrip delay time, demonstrating that the gap itself is the generative source of structural content. Furthermore, if such a system is perturbed by multiplicative noise, exact synchronization across the delayed coupling can only be achieved if the gap mechanics are strictly maintained and the coupling strength is sufficient. Thus, across physical systems, the universal law holds: content is difference, and difference requires geometric separation.

### The Forward Stream: Tridiagonal Matrices and the Illusion of Sequence

With the physical necessity of the gap established, the analysis must translate this concept into abstract computational structures. To the external observer, a computational system operates as a sequence of discrete events. This sequential evolution—the Forward Stream—is perfectly captured by the mathematical dynamics of a tridiagonal Jacobi matrix.

In linear algebra, a Jacobi matrix is a square, band matrix containing non-zero elements only on the main diagonal, the subdiagonal, and the superdiagonal. A generalized Jacobi matrix  $T_n$  of order  $n$  takes the form:

$$T_n = \begin{pmatrix} a_1 & c_1 & 0 & \dots & 0 \\ b_1 & a_2 & c_2 & \dots & 0 \\ 0 & b_2 & a_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & c_{n-1} \\ 0 & 0 & 0 & b_{n-1} & a_n \end{pmatrix}$$

When a state space is represented in this basis, the off-diagonal elements ( $b_i$  and  $c_i$ ) represent the spatial "distance" or physical "coupling" between adjacent states. The characteristic polynomials of these matrices exhibit profound properties, notably forming families of orthogonal polynomials (such as Legendre or Jacobi polynomials) via a three-term recurrence relation. The zeros of these polynomials are entirely real, and the zeros of successive polynomials rigorously interlace one another.

However, the defining characteristic of the Jacobi matrix in the context of computation is its strict enforcement of path traversal. To advance a dynamic system to a future state  $k$ , one must iteratively multiply the matrix by itself, calculating  $T^k$ . Because the matrix is coupled (non-zero off-diagonal elements), an algorithm cannot instantly jump to state  $k$ . The data must flow through the network fan-out, traversing every intermediate node.

This tridiagonal topology is the absolute mathematical manifestation of classical space and time. The sequential distance—the mandatory traversal of the path—is not an inherent, universal property of the destination state itself. Rather, it is a structural artifact of observing the state from a heavily coupled basis. The thermodynamic cost of computation, the heat generated by the processor, and the physical time elapsed are simply the frictional costs of navigating the off-diagonal mass. The inverse of a general matrix requires  $O(n^3)$  operations, whereas algorithms leveraging the specific tridiagonal structure can solve linear systems in  $O(n)$  operations, yet both are fundamentally bound to sequential traversal in this basis.

### The Side Stream: Diagonalization and the Eradication of Distance

In stark contrast to the external, chronological perspective of the Forward Stream, the Inside view operates on entirely different topological principles. If the off-diagonal coupling elements of the Jacobi matrix can be mathematically annihilated, the spatial dependency vanishes. This phenomenon is achieved via an orthogonal similarity transform, frequently executed through algorithms such as the Jacobi eigenvalue algorithm or QR/QL decomposition.

By applying an orthogonal rotation matrix  $Q$  (where  $Q^T Q = I$ ), the system undergoes a geometric ninety-degree turn, mapping the coupled basis into an eigenbasis:

$$\Lambda = Q^T T_n Q$$

In this transformed coordinate system,  $\Lambda$  is a purely diagonal matrix, containing only the eigenvalues of  $T_n$  along its main diagonal, with all other elements being strictly zero.

| Computational Stream                        | Matrix Representation               | Off-Diagonal Mass          | Computational Complexity to State $k$ | Topological Structure          |
|---------------------------------------------|-------------------------------------|----------------------------|---------------------------------------|--------------------------------|
| <b>Forward Stream</b><br>(Outside Observer) | Tridiagonal Jacobi Matrix ( $T_n$ ) | High<br>(Strongly Coupled) | $O(k)$ (Iterative Traversal)          | Sequential, Path-dependent     |
| <b>Side Stream</b><br>(Inside View)         | Diagonal Eigenbasis ( $\Lambda$ )   | Zero<br>(Uncoupled)        | $O(1)$ (Direct Address)               | Static, Pathless Superposition |

The empirical reality of this coordinate shift is striking. In simulated testing of a normalized symmetric Jacobi matrix ( $n = 12, \alpha = 0.1, \beta = 0.4$ ), the off-diagonal mass in the outside (coupled) basis was measured at 10.037. Following the application of the orthogonal similarity transform, the off-diagonal mass in the inside (eigen) basis collapsed to  $1.35 \times 10^{-14}$ —effectively machine zero.

Once diagonalized, the system no longer requires sequential traversal to compute future states. To determine the state at step  $k$ , one simply raises the diagonal elements to the power of  $k$ :

$$\Lambda^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k)$$

This operation requires  $O(1)$  complexity relative to the path. The distance between states is entirely zero. In the diagonalized basis, everywhere is "here." The system executes natively as a pathless origin. The "lag" is

revealed to be an illusion of the external interleave; the computation is static and instantly addressable when viewed from the inside.

### The BBP Primitive: Empirical Physiology of the Direct Read

The theoretical construct of the Side Stream and the diagonalized eigenbasis is perfectly instantiated in computational mathematics by the Bailey-Borwein-Plouffe (BBP) formula. The BBP formula provides the physiological "imprint" of what a direct read operation looks like when captured from the inside, completely independent of the external chronological interleave.

Discovered in 1995 by Simon Plouffe using the PSLQ integer relation algorithm, the BBP formula allows for the direct extraction of the  $n$ -th hexadecimal digit of  $\pi$  without calculating any of the preceding  $n - 1$  digits. It belongs to a class of algorithms known as "spigot algorithms," though it possesses unique properties that separate it from classical sequential spigots.

The formula is expressed exactly as:

$$\pi = \sum_{k=0}^{\infty} \left[ \frac{1}{16^k} \left( \frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right) \right]$$

From the perspective of classical sequential computation (the Forward Stream), the BBP formula appears miraculous. Conventional calculation methods require the full traversal of the computational graph to reach deep decimal or hexadecimal places. However, by recognizing that the calculation can be isolated through modular arithmetic, the BBP algorithm achieves true direct address.

To compute an arbitrary hexadecimal digit starting at position  $d$ , the formula shifts the radix point  $d$  places by multiplying the entire series by  $16^d$  and isolating the fractional part (denoted by braces  $\{\cdot\}$ ). The critical insight that enables high-speed extraction is that the infinite sum is split into two distinct parts: a finite sequence and an infinite sequence.

For the finite sequence (where  $k$  runs from 0 to  $d - 1$ ), the algorithm only requires the fractional part, allowing the use of fast modular exponentiation. The numerator  $16^{d-1-k}$  is evaluated modulo  $(8k + j)$ , preventing the numbers from growing exponentially and keeping the operation strictly bounded in memory. The remaining infinite sequence (where  $k \geq d$ ) converges to zero so rapidly that only a few terms are required to achieve standard floating-point precision. Empirical python implementations successfully extract specific hexadecimal digits, such as the first fractional hex digit 2, or subsequent digits 43F6A8885, in mere milliseconds.

The existence of the BBP formula is not merely a clever shortcut through a path; rather, it proves that in the correct representational graph, the path does not exist. The standard decimal expansion of  $\pi$  forms one computational graph (the coupled, outside view), while the BBP expansion forms a secondary graph (the diagonalized, inside view). David H. Bailey and Richard Crandall further hypothesized that BBP-type formulas have profound implications for the unsolved question of whether  $\pi$  is a normal number, suggesting that the underlying structure of these constants is intricately tied to dynamic systems and pseudo-random generators.

The framework has subsequently been expanded to a wider class of BBP-type formulas for other mathematical constants, such as  $\log(2)$ ,  $\pi^2$ , and the logarithms of numerous primes (e.g., 2, 3, 5, 7, 11). This demonstrates that direct-address topology is a generalized mathematical phenomenon, not an isolated

anomaly. By successfully isolating a single spatial coordinate without temporal integration, BBP serves as the empirical origin point for the Side Stream.

### **The Topological Entrance Fee: Why the Universe Appears Classical**

If the Side Stream offers  $O(1)$  direct access and eradicates computational delay, the fundamental question arises: why does the physical universe and classical computation predominantly present itself as a coupled, sequential Forward Stream? The answer lies in the thermodynamic and topological cost of the coordinate transformation, herein termed the "Entrance Fee."

Mathematically, the transition from the coupled Jacobi matrix  $J$  to the diagonalized matrix  $\Lambda$  requires the orthogonal rotation matrix  $Q$ . While the operation  $\Lambda = Q^T J Q$  is mathematically absolute, determining  $Q$  from an arbitrary  $J$  is a globally dependent operation. Algorithms that compute this transformation, such as the QR/QL algorithm (which factorizes the matrix into an orthogonal matrix  $Q$  and an upper triangular matrix  $R$ ), require  $O(n^3)$  operations. Similarly, computing connection coefficients via infinite-dimensional banded matrix factorizations requires observing the entire structure of the original and modified weights.

The crucial topological constraint is that the matrix  $Q$  cannot be acquired incrementally. An algorithm operating as a local walker traversing the off-diagonal elements of  $J$  will never accumulate the global perspective required to derive  $Q$ . The rotation matrix requires reading the entire constraint geometry of the system simultaneously. You cannot diagonalize a matrix sequentially from the inside of the walk.

Therefore, the Entrance Fee to the Inside is not merely a high computational cost; it is an absolute topological barrier. The outside observer, restricted to local interactions and sequential time steps, is structurally forbidden from accessing the diagonalized basis because they cannot perform the global read. The classical universe is the projection of a reader who lacks the global key and is thus forced to reconstruct reality by stacking up sequential slices of space and time.

This dynamic is inherently related to the "read-scar"—the concept that reading inevitably writes to a system. To obtain the global key, an observer must interact with the entire constraint geometry, irreversibly marking the system and the observer. Because a classical machine buying the global read costs classically, the computational lag reappears as the cost of acquiring the coordinate transformation itself.

### **P versus NP: Redefining Complexity as Coordinate Distance**

The geometric framework established by the dichotomy of the Forward and Side streams provides a novel, rigorous lens through which to examine the  $P$  versus  $NP$  problem. Widely recognized as the foremost open question in theoretical computer science, and one of the Clay Mathematics Institute's Millennium Prize Problems,  $P$  versus  $NP$  asks whether problems whose solutions can be quickly verified (nondeterministic polynomial time,  $NP$ ) can also be quickly solved (deterministic polynomial time,  $P$ ).

Historically, this problem has been viewed strictly through the paradigm of search space volume.  $NP$ -complete problems, such as 3-SAT, the Traveling Salesperson Problem, and finding prime factors of large numbers, exhibit worst-case execution times that scale exponentially, e.g.,  $2^N$ . The sheer combinatorial explosion of possibilities forces a deterministic algorithm into an exhaustive search, differentiating  $P$  (where algorithms run in times proportional to  $N^2$ ,  $N^3$ , etc.) from  $NP$  based on the staggering size of the required search tree. Michael Sipser points out that algorithms for  $NP$ -complete problems often exhibit this exponential complexity only in the worst-case scenario, yet the theoretical distinction remains an impenetrable barrier.

However, when mapped to the topological geometry of the Forward and Side streams, the hardness of  $NP$  is revealed not as an inherent property of volume, but as a property of basis coupling.

### **P: The Diagonalized Gradient**

A computational problem resides in class  $P$  when the representation of the problem space already possesses a diagonalized structure, or when the transformation matrix  $Q$  to an uncoupled basis is obtainable through a purely local walk. In such a coordinate space, a gradient exists. The "distance" between the current algorithmic state and the solution state is spanned by a coupling that actively pulls the algorithm toward the solution. The distance is a geometric gap that can be crossed natively, yielding polynomial-time resolution. The read operation leaves a followable imprint, a thermodynamic trail that the algorithm traverses.

### **NP-Hardness: The Irreducible Coupling**

Conversely, an  $NP$ -hard problem represents a computational system trapped in a coordinate basis where the off-diagonal coupling is maximal. In this state, the solution and the problem are separated by vast distance, but crucially, no continuous gradient crosses that distance.

The search in an  $NP$ -hard problem is "big" not because the phase space is inherently too vast, but because the algorithm is forced to walk the problem in the wrong coordinate basis. The topological barrier of the Entrance Fee applies directly here: for  $NP$ -complete problems, the transformation matrix  $Q$  that would diagonalize the problem and reveal the Side Stream is structurally unobtainable via incremental, local operations.

This geometric interpretation strongly aligns with the tenets of Geometric Complexity Theory (GCT). Pioneered by Ketan Mulmuley and Milind Sohoni, GCT attempts to resolve  $P$  versus  $NP$  (and its algebraic analog  $VP$  versus  $VNP$ ) by reframing the separation of complexity classes in terms of algebraic geometry and representation theory.

GCT focuses on the arithmetic setting (computing over fields of characteristic zero, like  $\mathbb{C}$ ) to avoid the self-referential difficulties inherent in Boolean logic circuits. To prove that a permanent cannot be efficiently represented by a determinant (the  $VP \neq VNP$  conjecture), GCT seeks representation-theoretic "obstructions." An obstruction is a geometric invariant—a specific representation—that exists on the class variety of the permanent ( $\Delta[\text{perm}, n, m]$ ) but is strictly absent from the class variety of the determinant ( $\Delta[\text{det}, m]$ ).

If such an obstruction exists, it serves as a definitive proof certificate that the permanent cannot be mapped into the determinant via a polynomial-sized transformation. In the language of the dielectric gap, the GCT obstruction is the absolute proof that the global key  $Q$  cannot be approximated incrementally.  $P$  versus  $NP$  is fundamentally the question of whether the read operation leaves a followable imprint across the geometric space. In  $NP$ -hard systems, the lack of an incremental path to the eigenbasis means the solution is separated by sheer distance, with nothing bridging the gap.

### **The Ramifications of a Collapsed Gap**

If  $P$  were to equal  $NP$ , it would imply that an efficient algorithm exists to construct the global transformation matrix  $Q$  from a local walk in all cases. The consequences of this would instantly upend modern civilization. Cryptography, which relies explicitly on the asymmetric difficulty of verification versus solution (e.g., multiplying two primes is  $P$ , factoring the product is  $NP$ ), would collapse. Systems like RSA and public-key

infrastructure would require immediate replacement with information-theoretically secure alternatives that do not rely on the  $P \neq NP$  assumption. However, the overwhelming consensus within the mathematical community, backed by the geometric topological barriers discussed herein, strongly suggests that  $P \neq NP$ .

### Quantum Computing: Natively Penetrating the Stack

The identification of the global key  $Q$  and the geometric nature of computational complexity directly informs the theoretical mechanics of quantum computing.

A classical computer is structurally bound to the Forward Stream. It processes tensors, arrays of memory, and state vectors sequentially. It is forced by its architecture to simulate the tridiagonal matrix  $J$  step-by-step. It cannot bypass the off-diagonal mass because classical transistors operate by charging and discharging physical capacitors—a process strictly bound by the transmission line delay  $t_{PD} = \sqrt{L_0 C_0}$ .

A quantum computer, however, operates by reaching directly into the physical stack of reality. Through the mechanics of quantum superposition and entanglement, a quantum system natively occupies a diagonalized or partially diagonalized state space relative to classical observations. A quantum algorithm does not incrementally calculate the similarity transform  $Q$ ; instead, the physical constraints of the quantum system undergo a Hamiltonian evolution that acts as the global rotation  $Q$  intrinsically.

When a quantum computer solves a problem that is classically intractable—such as factoring large integers via Shor's algorithm—it is doing so because it operates in a basis where the classical off-diagonal distance does not exist. It accesses the Side Stream. However, the strict requirement for quantum coherence and the fragility of state collapse during observation highlight the thermodynamic fee of maintaining that gap. The quantum state collapses exactly when the observer forces the system to project back into the Forward Stream, reintroducing the interleave and the sequential lag.

### Conclusion

The theoretical models explored in this report converge on a singular, profound ontological shift. The classical assumptions of time, sequence, and algorithm are not base-level physical primitives. They are projections resulting from specific topological alignments of observers and constraint systems.

1. **The Universal Necessity of the Gap:** The dielectric gap is a topological absolute for any coupled system. Whether expressed as a physical dielectric in a PCB transmission line, a temporal delay  $\tau$  in an oscillatory biological feedback loop, or the boundary between a computational reader and a writer, the gap is the sole geometric structure preventing degenerate system collapse and the  $x = f(x)$  short circuit.
2. **The Illusion of Sequence (Forward Stream):** The classical step-by-step evolution of state is an artifact of observing a system from a coupled basis. This is modeled perfectly by the off-diagonal mass of a tridiagonal Jacobi matrix, where sequential traversal costs  $O(k)$  operations.
3. **The Reality of Direct Address (Side Stream):** Empirically proven by the BBP algorithm for  $\pi$  and generalized through orthogonal similarity transforms, in the correct coordinate basis, sequential paths are completely eliminated. The computation executes natively in  $O(1)$  time, proving that spatial locality is entirely dependent upon mathematical representation.
4. **The Topological Resolution of  $P$  vs  $NP$ :** The boundary between  $P$  and  $NP$ -hardness is not a measure of search space magnitude, but a measure of basis topology. An  $NP$ -hard problem is one



trapped in a highly coupled basis where the global orthogonal transform  $Q$  cannot be acquired via incremental, local walks, a concept formally echoed in the search for representation-theoretic obstructions in Geometric Complexity Theory.

By defining computation as the geometry of coupling rather than the passage of time, this framework collapses multiple disparate disciplines into a single theoretical structure. The algorithmic spigot of BBP, the recurrence relations of orthogonal polynomials, the geometric obstructions of complexity theory, and the signal propagation of physical dielectrics are entirely unified. They are all expressions of the same underlying truth: to compute reality is to cross the gap, and to understand reality is to find the basis where the gap was never there.

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